

Jerome, Yat Hei Siu

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📅 Available to work immediately

🇬🇧 Right to work in the UK

🔗 jeromeehh.github.io/Jerome-Portfolio/

Profile

MSc Cybersecurity graduate (University of Birmingham) with a First-Class BSc in Computer Science and hands-on offensive security experience across RF, network, and IoT attack surfaces. Comfortable across the full penetration testing workflow, from reconnaissance and exploitation to vulnerability assessment and reporting, and quick to pick up new tools and techniques on the job. Full right to work in the UK without sponsorship, available immediately, and looking to bring this technical grounding and appetite for learning to a junior penetration testing or security analyst role.

Key Skills

- Programming & Scripting: Python, Java, Bash
- Application and Web Development in HTML, CSS, JavaScript
- Linux & Windows Command-Line Environments, incl. SSH-based system and forensic analysis
- Agile Software Development Methodologies
- Penetration Testing & Vulnerability Assessment
- Security Incident Root-Cause Analysis & Threat Modelling
- Blockchain & Decentralized Systems: Solidity smart contracts, Web3.js, Ethereum, Ganache
- Security Tools: Nmap, Metasploit, Wireshark, Bettercap etc
- RF & Hardware Security: SDR-based signal capture, demodulation
- Advanced Microsoft Office Environment Capabilities and Fast and accurate English typing (100 WPM)

Education

09/2024 - 01/2026 Msc Cyber Security, University of Birmingham

09/2022 - 07/2024 Bsc (Hons), Computer Science, University of Sussex - First Class Honours

Key Recent Projects

Security Analysis of Automotive Remote Keyless Entry (RKE) Systems:

- Engineered a black-box security analysis of a modern Mini Cooper RKE system using Software-Defined Radio (SDR) hardware to capture and reverse-engineer RF communications.
- Analysed and Demodulated FSK signals and decode Manchester II data streams.
- Executed a replay attack that bypassed a 48-bit rolling code system, uncovering a synchronization vulnerability and a denial-of-service (DoS) anti-replay lockout mechanism.

Network Exploitation & Device Reverse Engineering on Apple TV:

- Analysed wireless network traffic in monitor mode using Wireshark to investigate WPA encryption, and attempted Man-in-the-Middle (MitM) attacks utilizing Bettercap.
- Performed root-level filesystem analysis by establishing an SSH connection into a jailbroken Apple TV 4, subsequently extracting and reverse-engineering a third-party jailbreak application.

Blockchain-Integrated Smart Home IoT Architecture:

- Architected a decentralized smart home network utilizing Ethereum smart contracts (Solidity) to ensure tamper-proof data integrity and security for connected IoT sensors.
- Developed a Python-based MQTT-Smart Contract bridge and a lightweight Mosquitto broker to facilitate real-time, low-bandwidth communication between the blockchain network and edge devices.
- Deployed a locally simulated blockchain environment using Ganache and created a dynamic web dashboard utilizing Web3.js for secure user authentication and live device telemetry.

References:

Dr. Marius Muench, University of Birmingham

m.muench@bham.ac.uk

Prof. Martin White, University of Sussex

m.white@sussex.ac.uk

Additional Information

Mobility: Full UK Driving License. (Willing to commute/relocate)

Languages: English (Fluent) IELTS 7.0, Cantonese (Native), Mandarin (Fluent).

Links

GitHub: github.com/jeroomeehh

Portfolio: jeroomeehh.github.io/Jerome-Portfolio/

LinkedIn: linkedin.com/in/jeromesiu/